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> ML/LLM Inference Engineer — owns the full lifecycle of LLM inference optimization.

Inference optimization — Responsible for LLM serving optimization and achieving high reliability through researching new algorithms, modeling and validating their impact, and deploying them to customers. My tasks include choosing the best DP/TP strategy by modeling 10k-node inference clusters, tuning replica scheduling, and experimenting with different MoE EPLB placement strategies; reliability-aware scheduling using statistical approaches and discrete-event simulations; LLMOps methodologies for serving and deploying optimization code and supporting other LLM quantization research projects.

Predictive reliability — Responsible for failure forecasting and availability evaluation (Weibull modeling) of inference-cluster hardware (memory, NPUs, network) and software (vLLM, VeRL, agent workflows).

CAREER

Huawei — LLM Inference R&D Engineer · *September 2024 – present*

- Developed topology-aware combinatorial EPLB MoE expert-placement algorithms that co-optimize expert-to-NPU mapping with all-to-all communication and fused kernels to minimize network congestion and tail latency — up to 10% lower network imbalance and 5% TPOT improvement (95% CI ±1%).
- LLMOps for an LLM quantization project: owned the full lifecycle from research notebook → quantized model → deployed service and benchmarks, speeding up the research cycle by 3× (PyTorch, vLLM, DeepEval).
- Disaggregated Prefill/Decode simulator — Python discrete-event simulation of a prefill/decode serving system with KV-cache-aware load balancing and a Markov request-traffic generator; selects TP/DP/SP/EP topology, scheduling, and fault-recovery policy to hold 99.99% SLO. The same simulator was used to test the RL (VeRL) rollout-worker recovery strategy, cutting recovery time from 10 to 5 minutes and keeping the TPOT drop under failure scenarios to no more than 5%.
- NUMA-aware stress testing reached 95% of theoretical bandwidth; Weibull MTTR modeling + XGBoost failure forecasting cut repair costs by 2×. doc
- Time-series clustering of optical-power telemetry flags degrading links early; a minimal-parameter model (1M params) that runs on small chips raised cluster availability by +10% (no signal-to-noise requirement — raw optical-sensor data only). Used sktime for statistical and ML time-series analysis and PyTorch for clustering and pattern discovery.

ML Engineer — ITMO University · *November 2023 – September 2024*

- Sber & ITMO: CTGAN user emulator; improved BiVAE recommendation MAP@5 by 3%.
- Almazov & ITMO: Shapley-valued weighted XGBoost for obesity treatment; 71% accuracy for 3-month sibutramine therapy response; 2 Q2 papers. J. Pers. Med. J. Clin. Med.

- Industrial AI assistant: improved PDF NER with prompt engineering/self-consistency; filtering removed 80% repetition.

Testing Engineer — Harman · *September 2021 – July 2022*

- Sped up testing for complex GPS automotive systems with Robot Framework

Release Engineer — Intel · *July 2019 – August 2021*

- Supported Intel OneAPI HPC release infrastructure: Python/Bash automation and installer.

TECH STACK

- **AI infrastructure:** Harness, Agents workflow, RAG, CAG, vLLM, VeRL, PyTorch, Kubernetes
- **Data Analysis & Research:** Python, Pandas, Scikit-learn, AutoML, PatentScope, Elsevier Scopus

EDUCATION

ITMO University, 2024 — Master's degree in Data Analysis

– Thesis: Shapley-value weighted filtering for small-sample ML; +7% accuracy on medical data.

Higher School of Economics, 2022 — Bachelor's degree in Software Engineering

– Thesis: accessible assistant for visually impaired users using Yandex Cloud OCR.

PUBLICATIONS

- Google Scholar · h-index 2
- Danilov et al., J. Pers. Med. 2024, 14, 811 — sibutramine therapy prediction.
- Markina et al., J. Clin. Med. 2024, 13, 4151 — brown adipose tissue and obesity treatment.

COMPETITIONS

ScienceRAG — Research assistant agent based on GigaChat

– ScienceRAG: improved ArXiv RAG retrieval on GigaChat YouTube

X5 AI Tech Hackathon — Hallucination detection algorithm

– X5 AI Tech: ruBERT hallucination detector — 88% accuracy, <0.45 s inference; top-10.

LANGUAGES

- Russian, English (B1)